

Advanced Structural Analysis using ANSYS

Lecture and Exercise Notes

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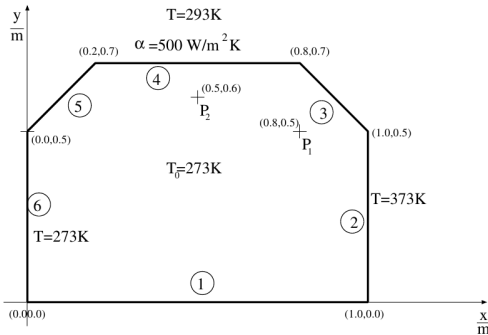
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Exercise 7: Thermal Analysis of Steel Body

Die Veranstaltung basiert u.a. auf dem Vorlesungs- und Übungsskript „Die Methode der finiten Elemente“, gehalten durch Prof. Dr. rer. nat. M. Braun, Dr.-Ing. H. Walter, Dr.-Ing. U. Zwiers und Dr.-Ing. Philipp Olle, am Lehrstuhl für Mechanik (Maschinenbau) der Universität Duisburg-Essen im WS2004/05

Thermal Analysis of a Steel Body

Problem description



- Objective target

distribution of temperature, gradient of temperature & heat flux in a steady state
(Find the needed amount of time with a matching number of time steps)

- Geometric data

see picture

- Material data

density $\rho = 7900 \text{ kg/m}^3$
 specific heat capacity $c = 490 \text{ J/(kg K)}$
 reference temperature $T_0 = 273 \text{ K}$
 thermal conductivity $\kappa = 47 \text{ W/(m K)}$

- Element types

2-D PLANE77

- Meshing

mapped mesh; varying mesh density

- Load/Boundary conditions

L 1,3,5: thermal isolation
 L 2,6: prescribed temp. values
 L 4: thermal transition
 (values see figure)

Additional ANSYS APDL Commands

Thermal Analysis of a Steel Body

Pre Processing

/PREP7

define table of temperature for
(temp. dependent) properties **MPTEMP**, ...
e.g. **MPTEMP**, 1, 100, 200, 300

setting material property data **MPDATA**, ...
e.g. **MPDATA**, DENS, 1, 1, 7500 for 1st temperature defined by **MPTEMP**, here 100
MPDATA, DENS, 1, 2, 8000 for 2nd temperature defined by **MPTEMP**, here 200
MPDATA, DENS, 1, 3, 8200 for 3rd temperature defined by **MPTEMP**, here 300

concatenates multiple lines **LCCAT**, ... (for mapped meshing)

free or mapped meshing **MSHKEY**, ...

Solution

/SOLU

analysis type **ANTYPE**, ... (time depending simulations by analysis type TRANS or 4)

prescribe temperature (INPUT: temperature) **DL**, ... (see manual for detail syntax)

prescribe transition **SFL**, ... (see manual for detail syntax)
(INPUT: heat transition coefficient (film coefficient) α and room temperature (bulk temperature))

assigns a uniform temperature to all nodes **TUNIF**, ... (in this example not used)

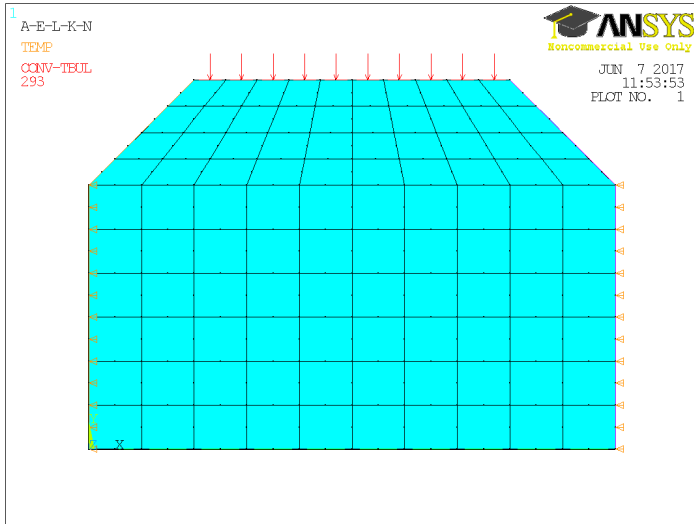
reference temperature for thermal strain calculations **TREF**, ... (in this example not used)

(NOTE: Initial Temperature is also set, if **TUNIF** is not used)
define size of (first) time step **DELTIM**, ... (see manual for detail syntax)
(not to use with automatic time stepping!!)

more help: help,XXX, whereby XXX should be replace by command.

Boundary value problem

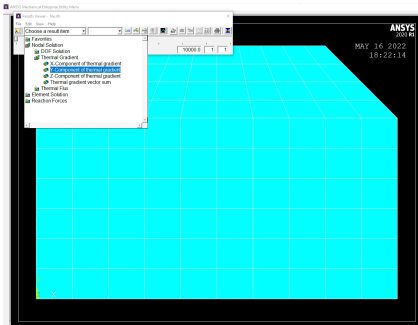
Thermal Analysis of a Steel Body



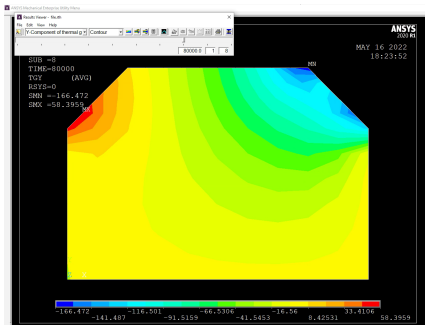
Time depending Post Processing in the graphical interface

In General Postproc → Result Viewer

choose result item



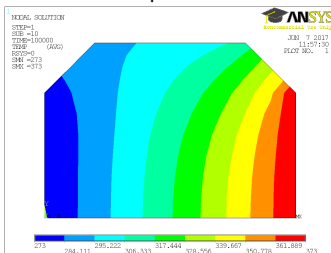
choose time



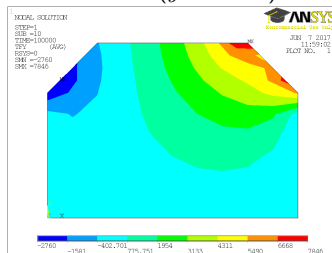
Results

Thermal Analysis of a Steel Body at steady state

temperature



heat flux (y -direction)



temperature gradient (in x -direction) temperature gradient (in y -direction)

